

```
[ubuntu@ip-172-31-12-254:~/devops-case$ kubectl get pods,svc,ingress
```

Dev install

NAME	READY	STATUS	RESTARTS	AGE
pod/app-6878c76-zwvbb	1/1	Running	0	27m

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/app	ClusterIP	10.109.201.52	<none>	80/TCP	27m
service/kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	25h

NAME	CLASS	HOSTS	ADDRESS	PORTS	AGE
ingress.networking.k8s.io/app	nginx	devops-case.dev.local	192.168.49.2	80	27m

```
[ubuntu@ip-172-31-12-254:~/devops-case$ kubectl describe pod -l app.kubernetes.io/instance=app | grep -E "Liveness|Readiness|Image:"
```

```
Image:          devops-case-app:0.1.0
```

```
Liveness:       http-get http://:http/healthz delay=5s timeout=2s period=10s #success=1 #failure=3
```

```
Readiness:      http-get http://:http/healthz delay=2s timeout=2s period=5s #success=1 #failure=2
```

```
[ubuntu@ip-172-31-12-254:~/devops-case$ helm list
```

NAME	NAMESPACE	REVISION	UPDATED	STATUS	CHART	APP VERSION
app	default	1	2026-05-22 09:40:21.199842458 +0000 UTC	deployed	app-0.1.0	0.1.0

```
ubuntu@ip-172-31-12-254:~/devops-case$ helm upgrade --install app charts/app \
-f charts/app/values-prod.yaml \
--set config.BUILD_SHA=$(git rev-parse --short HEAD) \
--set-string secret.DEMO_TOKEN=$(openssl rand -hex 16)
Release "app" has been upgraded. Happy Helming!
NAME: app
LAST DEPLOYED: Fri May 22 10:10:30 2026
NAMESPACE: default
STATUS: deployed
REVISION: 2
NOTES:
Release: app (app-0.1.0, app 0.1.0)
Namespace: default
Replicas: 2

Check rollout:
kubectl -n default rollout status deployment/app
kubectl -n default get pods -l app.kubernetes.io/instance=app

Ingress host(s):
http://devops-case.local/

To reach it from outside the cluster on minikube:
# one option: add the host to /etc/hosts pointing at $(minikube ip)
echo "$(minikube ip) devops-case.local" | sudo tee -a /etc/hosts
curl http://devops-case.local/ping
ubuntu@ip-172-31-12-254:~/devops-case$ kubectl rollout status deployment/app
deployment "app" successfully rolled out
ubuntu@ip-172-31-12-254:~/devops-case$ kubectl get pods,svc,ingress

NAME                                READY    STATUS    RESTARTS   AGE
pod/app-7789f4c6bf-c8wrj            1/1     Running   0           25s
pod/app-7789f4c6bf-tpf27            1/1     Running   0           23s

NAME                                TYPE                CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
service/app                        ClusterIP            10.109.201.52    <none>            80/TCP           30m
service/kubernetes                  ClusterIP            10.96.0.1        <none>            443/TCP          25h

NAME                                CLASS    HOSTS                ADDRESS          PORTS    AGE
ingress.networking.k8s.io/app      nginx    devops-case.local    192.168.49.2    80       30m
ubuntu@ip-172-31-12-254:~/devops-case$ kubectl get deployment app
NAME    READY    UP-TO-DATE    AVAILABLE    AGE
app     2/2      2             2            30m
```

Prod install

```
[ubuntu@ip-172-31-12-254:~/devops-case$ kubectl describe pod -l app.kubernetes.io/instance=app | grep -A2 -E "Limits|Requests"
```

```
Limits:
```

```
  cpu:      500m
```

```
  memory:   256Mi
```

```
Requests:
```

```
  cpu:      100m
```

```
  memory:   128Mi
```

```
--
```

```
Limits:
```

```
  cpu:      500m
```

```
  memory:   256Mi
```

```
Requests:
```

```
  cpu:      100m
```

```
  memory:   128Mi
```

Container CPU/memory requests & limits applied


```
ubuntu@ip-172-31-12-254:~/devops-case$ eval $(minikube docker-env)
ubuntu@ip-172-31-12-254:~/devops-case$ docker build --build-arg BUILD_SHA=$(git rev-parse --short HEAD) -t devops-case-app:0.1.1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  2.564MB
Step 1/14 : FROM python:3.12-slim AS builder
----> e1054bc5a9f2
Step 2/14 : WORKDIR /build
----> Using cache
----> e46679232939
Step 3/14 : COPY requirements.txt .
----> Using cache
----> fa277ba0ead9
Step 4/14 : RUN python -m venv /opt/venv && /opt/venv/bin/pip install --no-cache-dir --upgrade pip && /opt/venv/bin/pip install --no-cache-dir -r requirements.txt
----> Using cache
----> 892283176185
Step 5/14 : FROM python:3.12-slim
----> e1054bc5a9f2
Step 6/14 : RUN useradd --create-home --uid 1000 --shell /usr/sbin/nologin appuser
----> Using cache
----> 3f19ef2e33d8
Step 7/14 : COPY --from=builder --chown=appuser:appuser /opt/venv /opt/venv
----> Using cache
----> 8daae8188262
Step 8/14 : WORKDIR /app
----> Using cache
----> 83875cfbb83c
Step 9/14 : COPY --chown=appuser:appuser app.py .
----> Using cache
----> c8d6dd7dd410
Step 10/14 : USER appuser
----> Using cache
----> af3b5e87dd2a
Step 11/14 : ENV PATH=/opt/venv/bin:$PATH      PYTHONUNBUFFERED=1      PYTHONDONTWRITEBYTECODE=1      PORT=8080
----> Using cache
----> 2ff1dc08f056
Step 12/14 : EXPOSE 8080
----> Using cache
----> f92e20c35899
Step 13/14 : HEALTHCHECK --interval=30s --timeout=3s --start-period=5s --retries=3      CMD python -c "import urllib.request,sys; sys.exit(0 if urllib.request.urlopen('http://127.0.0.1:8080/healthz', timeout=2).status == 200 else 1)"
----> Using cache
----> 24c4e0c984c2
Step 14/14 : ENTRYPOINT ["gunicorn", "--bind", "0.0.0.0:8080", "--workers", "2", "--access-logfile", "-", "app:app"]
----> Using cache
----> b33ada767a6f
[Warning] One or more build-args [BUILD_SHA] were not consumed
Successfully built b33ada767a6f
Successfully tagged devops-case-app:0.1.1
ubuntu@ip-172-31-12-254:~/devops-case$ helm upgrade app charts/app -f charts/app/values-prod.yaml \
    --reuse-values --set image.tag=0.1.1
Release "app" has been upgraded. Happy Helming!
NAME: app
LAST DEPLOYED: Fri May 22 10:14:38 2026
NAMESPACE: default
STATUS: deployed
REVISION: 3
NOTES:
Release: app (app-0.1.0, app 0.1.0)
Namespace: default
Replicas: 2

[Check rollout:
  kubectl -n default rollout status deployment/app
  kubectl -n default get pods -l app.kubernetes.io/instance=app

Ingress host(s):
  http://devops-case.local/

To reach it from outside the cluster on minikube:
  # one option: add the host to /etc/hosts pointing at $(minikube ip)
  echo "$(minikube ip) devops-case.local" | sudo tee -a /etc/hosts
  curl http://devops-case.local/ping
ubuntu@ip-172-31-12-254:~/devops-case$ kubectl rollout status deployment/app
deployment "app" successfully rolled out
```

Rollout

```
ubuntu@ip-172-31-12-254:~/devops-case$ helm history app
REVISION      UPDATED              STATUS      CHART          APP VERSION      DESCRIPTION
1             Fri May 22 09:40:21 2026    superseded  app-0.1.0      0.1.0            Install complete
2             Fri May 22 10:10:30 2026    superseded  app-0.1.0      0.1.0            Upgrade complete
3             Fri May 22 10:14:38 2026    deployed   app-0.1.0      0.1.0            Upgrade complete
```

```
ubuntu@ip-172-31-12-254:~/devops-case$ helm rollback app
Rollback was a success! Happy Helming!
```

```
ubuntu@ip-172-31-12-254:~/devops-case$ kubectl rollout status deployment/app
deployment "app" successfully rolled out
```

```
ubuntu@ip-172-31-12-254:~/devops-case$ helm history app
[REVISION      UPDATED              STATUS      CHART          APP VERSION      DESCRIPTION
1             Fri May 22 09:40:21 2026    superseded  app-0.1.0      0.1.0            Install complete
2             Fri May 22 10:10:30 2026    superseded  app-0.1.0      0.1.0            Upgrade complete
3             Fri May 22 10:14:38 2026    superseded  app-0.1.0      0.1.0            Upgrade complete
4             Fri May 22 10:15:28 2026    deployed   app-0.1.0      0.1.0            Rollback to 2
```

Rollback to 2